Narration — final video script (EN) · v4 · 2026-10-26 (corrector-final, hallazgos QA)

> Generado por `slides/script/build_narration.py` desde `slides/script/narration_en_template.md` con `figures/numbers.json` (solo claves `verified: true`). No editar a mano: se cambia la plantilla y se regenera. Fuente única de la voz: `docs/00_storyboard_diapositivas.md` la copia palabra por palabra.
> 601 palabras (`wc -w` , cifra = 1) · 611 habladas · 17 cifras habladas · 294.2 s = 4:54.2 (150 ppm + 1,5 s de aire por cifra + 0,6 s por transición; pasos con MP4 ≥ MP4 + 0,3 s) · 32 pasos · 71 bloques de subtítulo.
> Gancho: la paradoja se plantea antes de 12.0 s y el giro "Yet 84 are still growing." empieza en 12.0 s.

Lectura: una frase por línea `>` ; leer sin prisa, ≈ 1,5 s de silencio después de cada cifra (marcadas en la tabla). Nombres: "UNALSta-R" = *u-nal-sta-ar*. Lo que no lleva cifra en voz queda en pantalla.

Tiempos por paso

Paso	Diapositiva	`data-fragment-index` (HTML → reveal)	Inicio–fin	Dur. (s)	Palabras	Cifras habladas	MP4
`cover.b1`	Cover	sección	0:00.0–0:02.7	2.7	3	1	—
`cover.b2`	Cover	2 → 0	0:02.7–0:12.0	9.3	15	1	—
`cover.b3`	Cover	3 → 1	0:12.0–0:25.1	13.1	20	3	—
`cover.b4`	Cover	4 → 2	0:25.1–0:27.7	2.6	5	0	—
`cover.b5`	Cover	5 → 3	0:27.7–0:29.9	2.2	4	0	—
`s1.b0`	Slide 1	sección	0:29.9–0:47.3	17.4	42	0	—
`s1.b1`	Slide 1	1 → 0	0:47.3–0:51.9	4.6	10	0	—
`s1.b2`	Slide 1	2 → 1	0:51.9–1:04.8	12.9	26	1	fig01_growth_components_by_income (12.1 s)
`s1.b3`	Slide 1	3 → 2	1:04.8–1:08.2	3.4	7	0	—
`s1.b3b`	Slide 1	4 → 3	1:08.2–1:14.3	6.1	10	1	—
`s1.b4a`	Slide 1	5 → 4	1:14.3–1:18.9	4.6	10	0	—
`s1.b4b`	Slide 1	6 → 5	1:18.9–1:22.3	3.4	7	0	—
`s1.b4c`	Slide 1	7 → 6	1:22.3–1:26.9	4.6	10	0	—
`s1.b5a`	Slide 1	8 → 7	1:26.9–1:31.2	4.3	7	0	fig02b_lever_map_zoom (4.0 s)
`s1.b5b`	Slide 1	9 → 8	1:31.2–1:39.7	8.5	15	1	—
`s2.b0`	Slide 2	sección	1:39.7–1:40.3	0.6	0	0	—
`s2.b1`	Slide 2	1 → 0	1:40.3–1:59.1	18.8	38	2	fig04_same_road_registration (18.5 s)
`s2.b2`	Slide 2	2 → 1	1:59.1–2:12.7	13.6	21	2	—
`s2.b3`	Slide 2	3 → 2	2:12.7–2:21.3	8.6	20	0	—
`s2.b4`	Slide 2	4 → 3	2:21.3–2:39.1	17.8	43	0	—
`s3.b0`	Slide 3	sección	2:39.1–2:39.7	0.6	0	0	—
`s3.b1`	Slide 3	1 → 0	2:39.7–2:54.9	15.2	29	1	fig07_single_indicator_scorecard (14.9 s)
`s3.b2`	Slide 3	2 → 1	2:54.9–3:01.8	6.9	12	1	—
`s3.b3`	Slide 3	3 → 2	3:01.8–3:12.5	10.7	23	0	fig09_pca_two_dimensions (10.4 s)
`s3.b4`	Slide 3	4 → 3	3:12.5–3:36.5	24.0	51	2	—
`s4.b0`	Slide 4	sección	3:36.5–3:37.1	0.6	0	0	—
`s4.b1`	Slide 4	1 → 0	3:37.1–4:05.6	28.5	66	1	fig10_trajectory_groups (28.2 s)
`s4.b2`	Slide 4	2 → 1	4:05.6–4:10.6	5.0	11	0	—
`s4.b3`	Slide 4	3 → 2	4:10.6–4:28.4	17.8	42	0	—
`s4.b4`	Slide 4	4 → 3	4:28.4–4:40.6	12.2	29	0	—

Paso	Diapositiva	`data-fragment-index` (HTML → reveal)	Inicio–fin	Dur. (s)	Palabras	Cifras habladas	MP4
`s4.b5`	Slide 4	5 → 4	4:40.6–4:51.2	10.6	25	0	—
`refs`	References	sección	4:51.2–4:54.2	3.0	0	0	—
Total			0:00.0–4:54.2	**294.2**	**601**	**17**	

Cover — One number can't tell 8 billion stories (0:00.0–0:29.9 · 47 palabras)

`cover.b1` · inicio de sección · 0:00.0–0:02.7

> **Eight billion people.**

`cover.b2` · fragmento 2 · 0:02.7–0:12.0

> **In 2023, 130 countries and areas had fertility below the level that replaces each generation.**

`cover.b3` · fragmento 3 · 0:12.0–0:25.1

> **Yet 84 are still growing.**
> **66 have more births than deaths; 18 grow only because more people arrive than leave.**

`cover.b4` · fragmento 4 · 0:25.1–0:27.7

> **One number can't explain that.**

`cover.b5` · fragmento 5 · 0:27.7–0:29.9

> **We are Team UNALSta-R.**

Slide 1 — Question 1: which lever moves each country's population? (0:29.9–1:39.7 · 144 palabras)

`s1.b0` · inicio de sección · 0:29.9–0:47.3

> **Three questions.**
> **One: how do births, deaths and migration relate, by region, country and income?**
> **Two: does one indicator tell the whole story, or only several together?**
> **Three: do countries share one road at different speeds, and what should the UN watch?**

`s1.b1` · fragmento 1 · 0:47.3–0:51.9

> **Population change is bookkeeping: births minus deaths, plus net migration.**

`s1.b2` · fragmento 2 · 0:51.9–1:04.8 · MP4 fig01_growth_components_by_income 12.1 s

> **In today's high-income countries, net migration accounted for about seven in ten of the growth since 2000.**
> **Elsewhere, growth came almost entirely from births exceeding deaths.**

`s1.b3` · fragmento 3 · 1:04.8–1:08.2

> **Blue: more births than deaths, and growing,**

`s1.b3b` · fragmento 4 · 1:08.2–1:14.3

> **including 22 below replacement that grow with no net arrivals.**

`s1.b4a` · fragmento 5 · 1:14.3–1:18.9

> **Light blue: deaths exceed births, but arrivals keep it growing.**

`s1.b4b` · fragmento 6 · 1:18.9–1:22.3

> **Orange: deaths exceed births, and it shrinks.**

`s1.b4c` · fragmento 7 · 1:22.3–1:26.9

> **Purple: births exceed deaths, but net departures make it shrink.**

`s1.b5a` · fragmento 8 · 1:26.9–1:31.2 · MP4 fig02b_lever_map_zoom 4.0 s

> **Migration is also the least certain lever.**

`s1.b5b` · fragmento 9 · 1:31.2–1:39.7

> **For the Emirates in 2050, the UN's 95 percent range is nine to thirty-three million.**

Slide 2 — Question 1 (cont.): same road, different speed (1:39.7–2:39.1 · 122 palabras)

`s2.b0` · inicio de sección · 1:39.7–1:40.3

(sin voz: transición)

`s2.b1` · fragmento 1 · 1:40.3–1:59.1 · MP4 fig04_same_road_registration 18.5 s

> **Set each country's clock to when its fertility fell below four children, and the curves line up.**
> **The road is shared; the speed is not.**
> **China went from five to three children in 5 years; Egypt took 33.**

`s2.b2` · fragmento 2 · 1:59.1–2:12.7

> **Survival analysis: child deaths started falling first in 106 of 109 high-fertility countries.**
> **The gap was typically at least 22 years.**

`s2.b3` · fragmento 3 · 2:12.7–2:21.3

> **A within-country panel regression finds this link smaller than comparing countries suggests.**
> **A sequence in time, not proof of cause.**

`s2.b4` · fragmento 4 · 2:21.3–2:39.1

> **Question one, answered: child deaths fell first and births followed, at each country's pace; latest in low-income countries.**
> **The link is clearest in sub-Saharan Africa.**
> **Migration moves on its own, alike among neighbours only in the Gulf, and weighs most in high-income countries.**

Slide 3 — Question 2: does one indicator tell the whole story? (2:39.1–3:36.5 · 115 palabras)

`s3.b0` · inicio de sección · 2:39.1–2:39.7

(sin voz: transición)

`s3.b1` · fragmento 1 · 2:39.7–2:54.9 · MP4 fig07_single_indicator_scorecard 14.9 s

> **Question two.**
> **Fertility alone captures the slow story: about three quarters of the differences in how old populations are.**
> **Only part of today's growth, and almost none of migration.**

`s3.b2` · fragmento 2 · 2:54.9–3:01.8

> **Fertility alone finds only one in eight of each country's closest look-alikes.**

`s3.b3` · fragmento 3 · 3:01.8–3:12.5 · MP4 fig09_pca_two_dimensions 10.4 s

> **Principal component analysis: countries differ in two separate ways.**
> **Fertility tracks the first, the transition to fewer births and deaths, but misses migration.**

`s3.b4` · fragmento 4 · 3:12.5–3:36.5

> **Blend indicators into one score: the weights can shift one country in five by more than twenty places.**
> **Question two, answered: no.**
> **The Emirates and Belarus share the same fertility; one grows almost four percent a year, the other shrinks.**
> **The story emerges only when transition and migration are read together.**

Slide 4 — Question 3: one road, four groups — and what to watch on each (3:36.5–4:51.2 · 173 palabras)

`s4.b0` · inicio de sección · 3:36.5–3:37.1

(sin voz: transición)

`s4.b1` · fragmento 1 · 3:37.1–4:05.6 · MP4 fig10_trajectory_groups 28.2 s

> **Question three.**
> **K-means groups countries' seventy-year paths into three stretches of one road, plus a migration-led group, mostly Gulf states.**
> **Early movers are projected to grow only through migration, or shrink.**
> **Fast movers, India, China and Indonesia, are projected to age from 30 to 40 in about 32 years.**
> **About as fast as Early movers did, or faster.**
> **Recent movers have the highest child mortality today.**

`s4.b2` · fragmento 2 · 4:05.6–4:10.6

> **Countries grew closer in life expectancy, but further apart in age.**

`s4.b3` · fragmento 3 · 4:10.6–4:28.4

> **Question three, answered: mostly one road, at different speeds.**
> **For the UN: watch child survival where it is off pace for the 2030 goal.**
> **Watch ageing where fertility is low but populations still grow.**
> **Watch migration data where growth depends on arrivals.**

`s4.b4` · fragmento 4 · 4:28.4–4:40.6

> **Following ASA guidelines, projections are scenarios; hypotheses were fixed before the final run.**
> **These are patterns, not causes; every number and our AI use are documented in our notebook.**

`s4.b5` · fragmento 5 · 4:40.6–4:51.2

> **One number can't tell eight billion stories.**
> **But births, deaths and migration, read together region by region, can tell the UN where to look next.**

References — References (4:51.2–4:54.2 · 0 palabras)

`refs` · inicio de sección · 4:51.2–4:54.2

(sin voz: referencias en pantalla 3 s)

Claves usadas en la voz

Clave	Valor formateado	say	verified
`cluster_examples_fast`	India, China and Indonesia	India, China and Indonesia	✔
`dur53_chn`	5	—	✔
`dur53_egy`	33	—	✔
`knn_tfr`	13	one in eight	✔
`lag_onset_lower_bound`	22	at least 22 years	✔
`n_below_repl_growing_2023`	84	—	✔
`n_below_repl_natural_growth_2023`	66	—	✔
`n_below_replacement_2023`	130	—	✔
`n_grow_only_migration_2023`	18	18	✔
`n_momentum_countries`	22	22	✔
`n_mort_first_onset`	106	—	✔
`n_onset`	109	—	✔
`pop_are_2050_pi95_high`	33.3	thirty-three	✔
`pop_are_2050_pi95_low`	8.9	nine	✔
`r2_tfr_medage`	0.77	three quarters	✔
`share_mig_high_2000_2023`	70.7	seven in ten	✔
`share_rank_mc20`	22	one country in five	✔
`team_name`	UNALSta-R	—	✔
`twin_growth_are`	+3.70	four	✔
`world_pop_2023_bn`	8.1	eight billion	✔
`yrs_medage30_40_recent`	32	about 32	✔